Task 3: Data Exploration Report

Validated Encoders

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1 Verify Annotations & Qualitative Analysis

We reviewed 10 audio recordings via Label Studio. For each recording, we compared annotations from multiple annotators, checked class assignments against class definitions, verified temporal precision of onsets and offsets, ensured the one-second rule was respected for pauses, and identified any missed events. Incorrect annotations were corrected where possible and ambiguous cases were commented and accepted.

1.1 Summary of Disagreements

The overall disagreement was mainly about temporal boundaries, specifically onsets and offsets. In many cases the one-second rule was violated, and in some cases target classes audible in the recording were neither annotated nor stated in the metadata. Disagreement does not always imply that annotators explicitly violated the rules. Short or faintly audible sounds, critical overlapping among events, and transitions between similar sources (e.g., toilet refilling and running water) were the most frequent causes of disagreement among annotators.

1.2 Case Study

Audio file 003679 lists *phone_ringing* and *light_switch* as target classes. Upon review, clearly audible footsteps are present throughout the recording but are neither mentioned in the metadata nor annotated by one of the annotator. Furthermore, the two annotators disagree on the *phone_ringing* boundaries: one annotator clearly violates the one-second rule by not splitting a phone ringing event that contains a pause longer than one second. Figure 1 shows the side-by-side comparison.

2 Annotations

2.1 Annotator Agreement

We quantified agreement using Cohen's Kappa and Jaccard similarity, computed pairwise across all annotator pairs in files with at least two annotators. Per-class kappa was computed only in files where at least one annotator marked the class as present, avoiding inflated scores from trivial zeroes agreements.

The overall mean Cohen's Kappa across 5,735 pairs is $\kappa = 0.788$ (std: 0.201), indicating **substantial agreement** on the Landis & Koch scale. All 15 classes achieve at least substantial agreement. The lowest agreement is for *footsteps* ($\kappa = 0.770$), *door_open_close* ($\kappa = 0.839$), and *light_switch* ($\kappa = 0.898$)—short events with ambiguous temporal boundaries. The highest agreement is for *coffee_machine* ($\kappa = 0.983$), *vacuum_cleaner* ($\kappa = 0.981$), and *bell_ringing* ($\kappa = 0.979$)—sustained, acoustically distinct sounds (see Figure 2).

The mean overlap ratio between annotated classes and the uploader's target class tags is 0.90. In 85.6% of files all target classes appear in the annotations, confirming that temporal annotations generally agree with uploader metadata. In 16.6% of files annotators identified additional classes beyond the targets, probably detected as incidental background events.

Pairs involving the recording owner show slightly higher agreement ($\kappa = 0.792$, $n = 4{,}170$) than pairs without ($\kappa = 0.776$, $n = 1{,}565$), suggesting first-hand knowledge of the recorded scene which provides a slight annotation advantage.

2.2 Convert Annotations to Labels

We convert the raw annotation array $[T, C, A]$ into binary labels $[T, C]$ in two steps. First, each annotator's soft overlap value is binarized at threshold of 0.5: values above indicate the event is present. Second, we take the majority vote by averaging binary decisions across annotators and thresholding at ≥ 0.5.

This assumes all annotators are equally reliable and that overlap > 0.5 constitutes presence. With two annotators (the most common case), a tie resolves as positive, introducing a slight bias toward labeling events as present. If all annotators make the same error, majority vote preserves it. Partial overlaps near 0.5 are sensitive to the exact threshold choice, e.g, 0.49 are not considered.

2.3 Label Characteristics

The dataset spans 168,239 segments ($\sim$46.7 hours). The most frequent class is *footsteps* (1,815 files, 15.3% of total time); the rarest is *bell_ringing* (276 files, 1.5%), revealing a significant class imbalance (see Table 1 and Figure 3).

Event durations vary by class type: sustained sounds have long durations (*vacuum_cleaner*: median 32s, *microwave*: 24s, *coffee_machine*: 19s), while transient events are short (*light_switch*: 2s, *door_open_close*: 3s). Approximately 7.3% of segments are silent where no active classes are found (silent or non target classes). *Footsteps* co-occur frequently with many other classes as expected.

3 Audio Features

3.1 Metadata Distribution

Recording properties were extracted from the *.npz files. 53.9% of device placement are static and 46.1% mobile. The most common recording devices are smartphones (mainly iPhones). The most frequent recording environments are *kitchen*, *living room*, and *office*, with strong environment-class coupling: kitchen recordings are dominated by *running_water* and *cutlery_dishes*, while office recordings feature *keyboard_typing* and *phone_ringing*.

3.2 Feature Statistics

The dataset contains 960 feature dimensions (14 base features $\times$ 4 aggregation methods). Feature scales vary dramatically: *power_max* reaches 67,941 while *flatness* stays in $[0, 1]$, confirming that standardization is essential before any distance based method.

Temporal derivative features (*mfcc_d_mean*, *mfcc_d2_mean*) have means values ranging near zero across all classes. Their *std* and *min/max* aggregations do not carry useful information. Energy and power are not normally distributed (mean $\ll$ max), explained by probable outliers or burst events (non target classes) which can be considered as bias.

3.3 Feature Correlation

We analyzed inter-group correlation using Pearson correlation on mean aggregated values. Strong correlations reflect known physical relationships: *zcr–centroid* ($r = 0.95$), *rolloff_high–bandwidth* ($r = 0.96$), *energy–flux* ($r = 0.93$). MFCCs and mel spectrogram correlate at $r = 0.82$. Temporal derivatives (*mfcc_d*, *mfcc_d2*) show near-zero correlation with all other features ($|r| < 0.14$).

The class-wise feature profile (Figure 11) reveals distinctive acoustic fingerprints. *Vacuum_cleaner* shows the highest energy (+1.77σ) and mel spectrogram (+1.35σ), reflecting loud broadband noise. *Running_water* has the highest zcr (+1.41σ) and flatness (+1.20σ). *Microwave* shows very low centroid (-1.02σ) and bandwidth (-1.24σ), for its low frequency nature and influnced by the buttons/ stat/end beeps. *Footsteps* and *keyboard_typing* share similar, below average profiles, that make them difficult to separate.

4 Feature Space Visualization

We selected spectral features (mel spectrogram, centroid, bandwidth, flatness, contrast, rolloff) with mean and std aggregations (280 dimensions), chosen based on our correlation analysis for complementary spectral coverage with minimal redundancy. PCA compresses these to 50 components retaining 97.8% of variance (only 9 components needed for 90%), substantially outperforming alternative feature subsets we tested.

We filtered for single class segments with $\geq$80% annotator agreement (76,716 segments), then subsampled 14,994 for t-SNE (perplexity=30, PCA initialization, 1,000 iterations). This achieved the lowest KL divergence (1.99) among all tested configurations. Figure 13 shows well and clearly separated clusters for sustained sounds (*vacuum_cleaner*, *running_water*) and more diffuse, overlapping distributions for transient events (*footsteps*, *light_switch*). Pairwise analysis confirms partial overlap between *running_water* and *toilet_flushing*, while *vacuum_cleaner* and *microwave* form distinct clusters despite both being sustained appliance sounds.

5 Conclusions

5.1 Biases

Three key biases were identified: (1) **Device bias**—a small number of smartphone models dominate the dataset, introducing device specific frequency responses and noise floors that classifiers may learn instead of sound event characteristics. (2) **Environment bias**—certain classes are strongly tied to specific rooms (e.g., *running_water* to kitchens), so a classifier may learn room acoustics rather than the sound event itself. (3) **Annotator bias**—with mostly two annotators per file, systematic errors directly affect labels; the recording owner annotates with slightly higher agreement and may introduce bias toward intended target sounds.

5.2 Dataset Suitability

The dataset is suitable for training sound event detectors, but several challenges exist. Class imbalance (*footsteps* appears in $9\times$ more files than *coffee_machine*) requires handling. Similar class pairs (*footsteps* vs. *keyboard_typing*, *door* vs. *wardrobe*) share overlapping feature profiles, making precise discrimination challenging. Feature scale differences require normalization. The high annotator agreement ($\kappa = 0.79$) provides generally reliable labels, though transient events may introduce label noise near temporal boundaries.

5.3 Broader Impact

A domestic sound dataset has many applications in areas such as elderly monitoring and smart home automation. However, it also presents risks of misuse, including unauthorized surveillance. To mitigate these concerns, dataset creators should ensure the anonymity of data contributors, limit redistribution and usage, and enforce strict ethical guidelines and agreements. Additionally, the dataset license should explicitly prohibit any use for surveillance purposes.

6 Figures and Tables

6.1 Figures

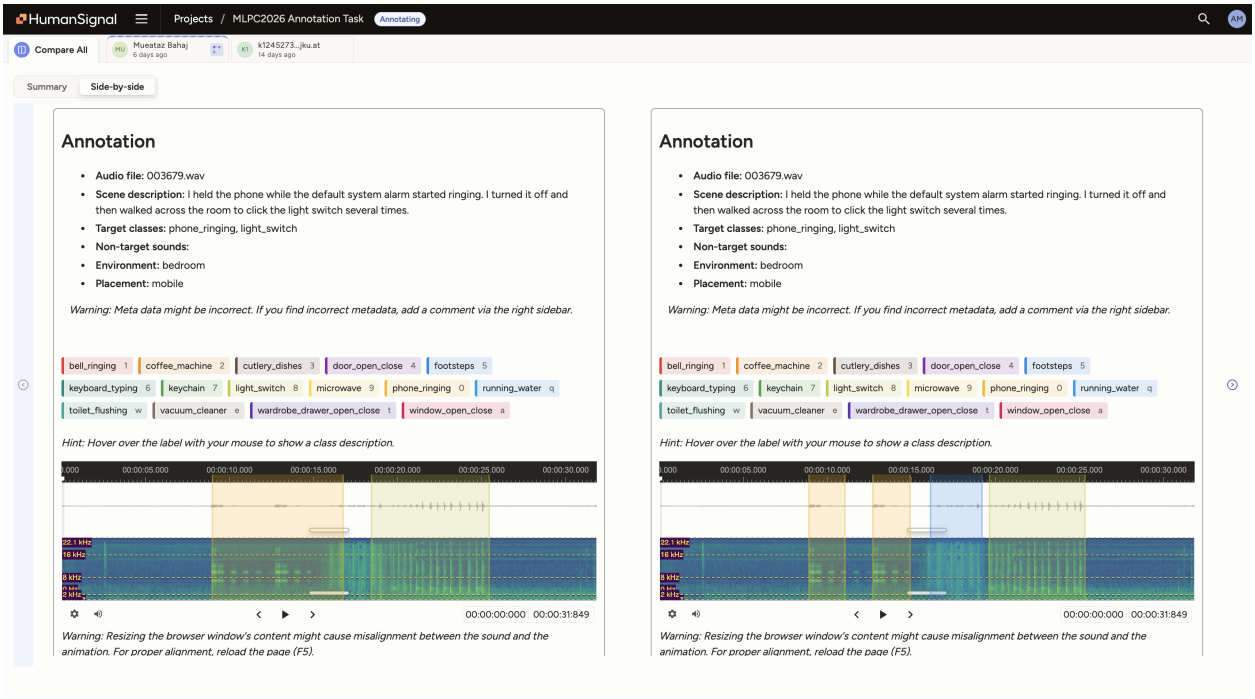


Figure 1: Side-by-side annotation comparison for file 003679. Annotator 1 (left) does not split the phone ringing region despite a pause exceeding one second. Also Footsteps are absent from the annotation.

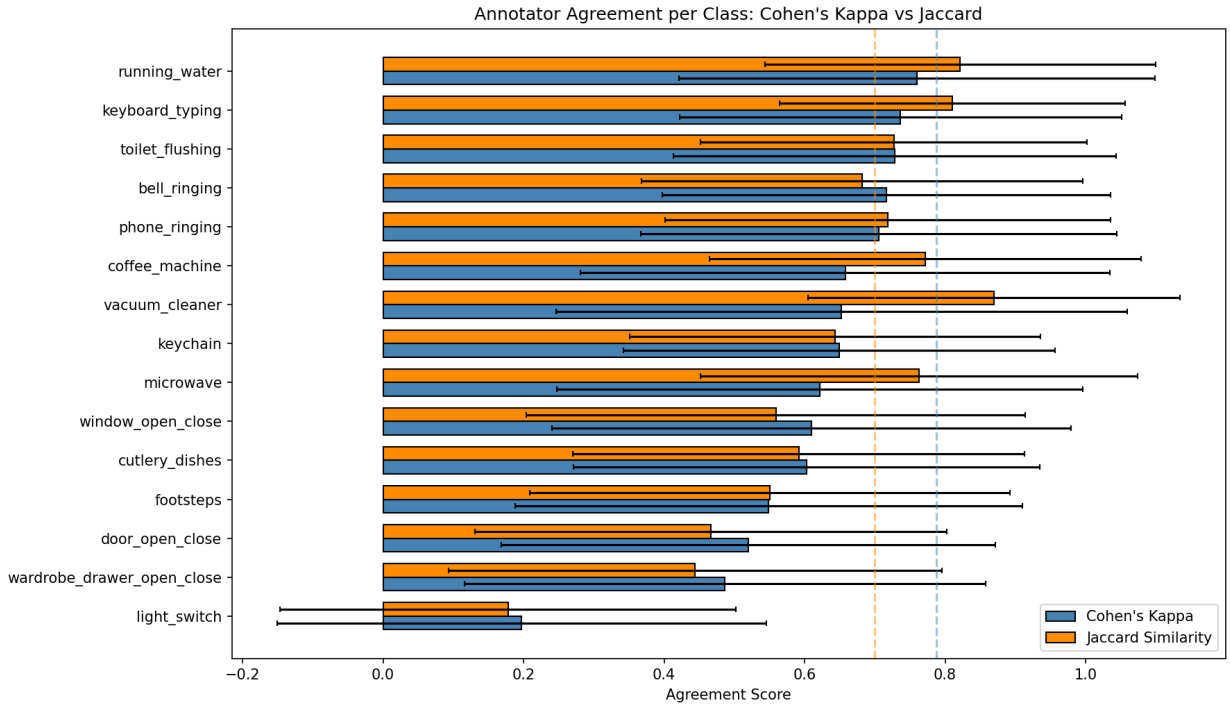


Figure 2: Class-wise annotator agreement: Cohen's Kappa vs. Jaccard similarity. Dashed lines indicate overall means. Transient events (footsteps, door) show lower agreement than sustained sounds (vacuum, coffee machine).

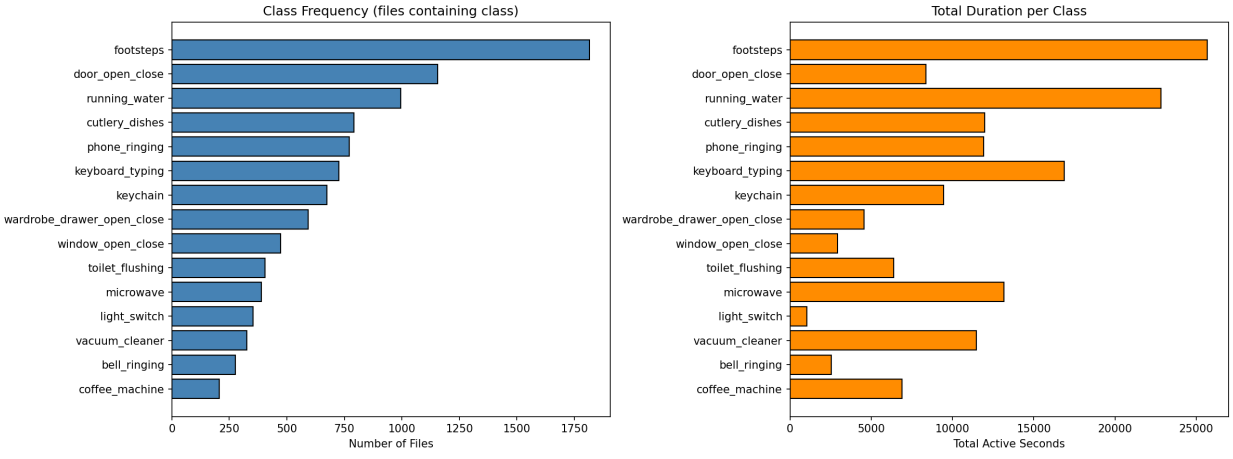


Figure 3: Left: number of files containing each class. Right: total active seconds per class.

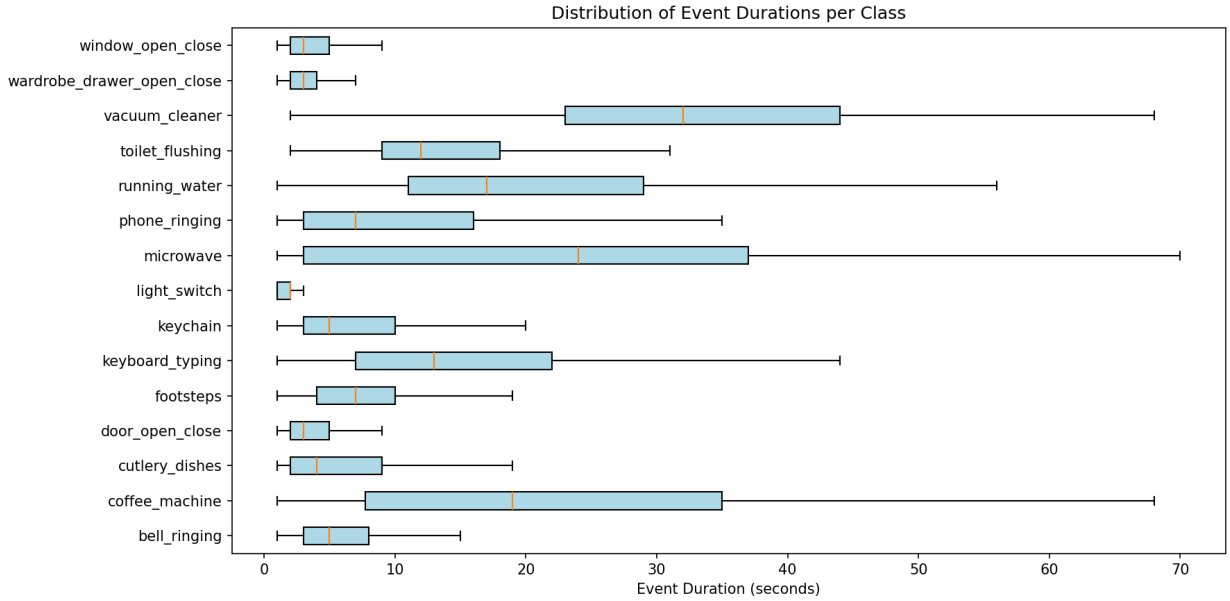


Figure 4: Event duration distribution per class (outliers hidden).

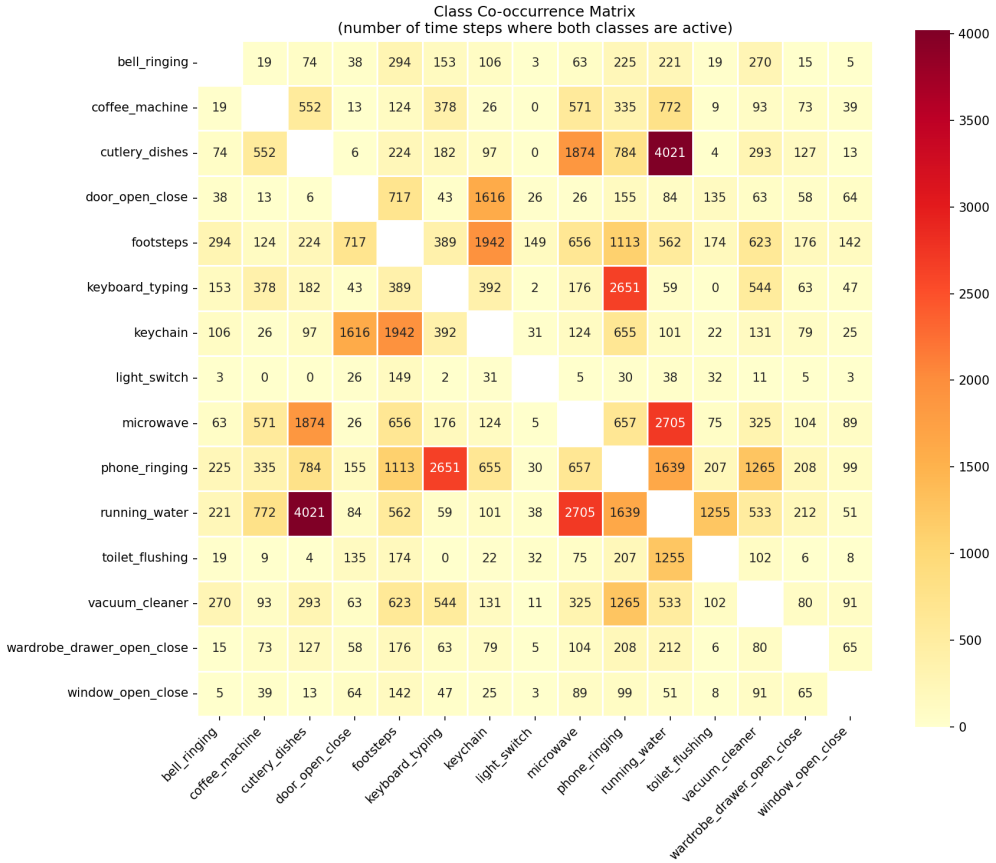


Figure 5: Class co-occurrence matrix: number of time steps where both classes are simultaneously active.

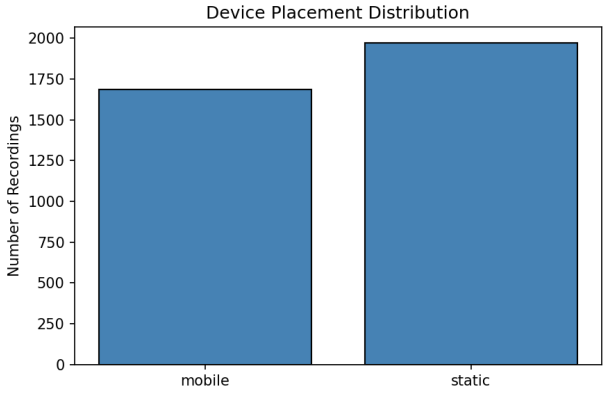


Figure 6: Distribution of device placement (static vs. mobile).

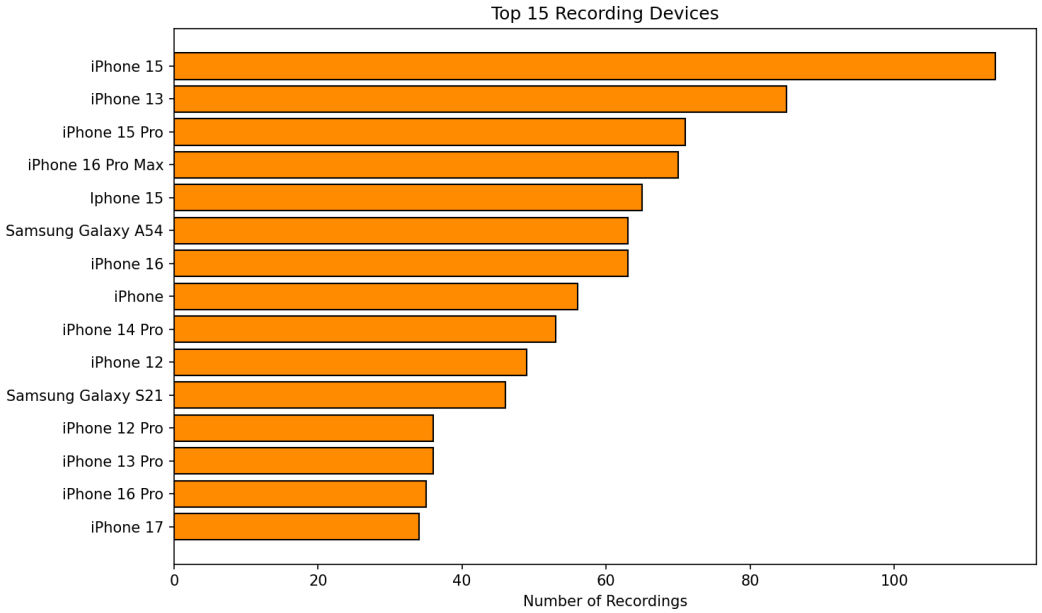


Figure 7: Top 15 recording devices. Iphone and Samsung smartphones dominate the dataset.

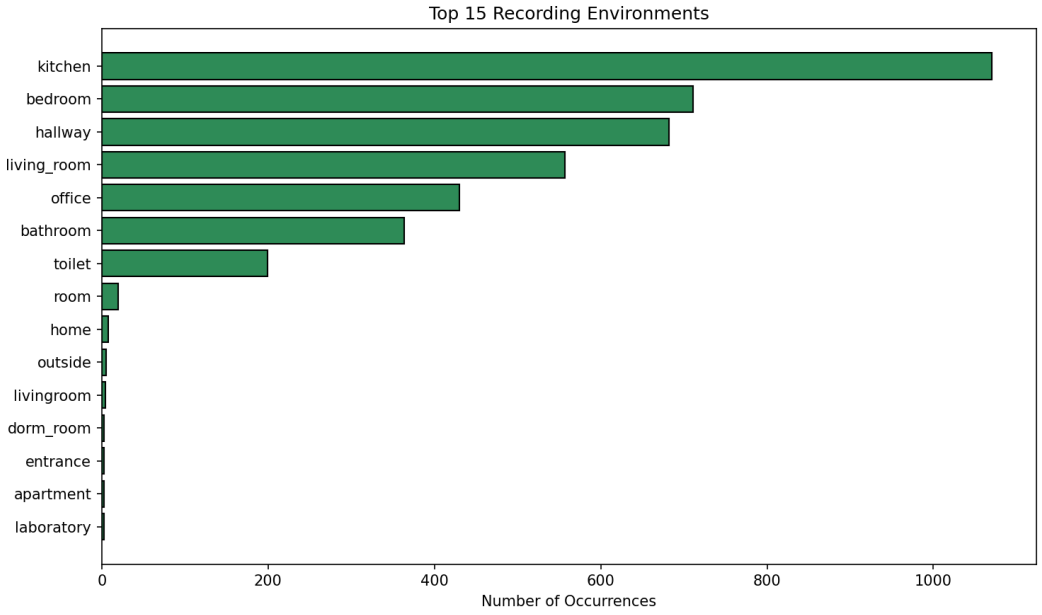


Figure 8: Top 15 recording environments.

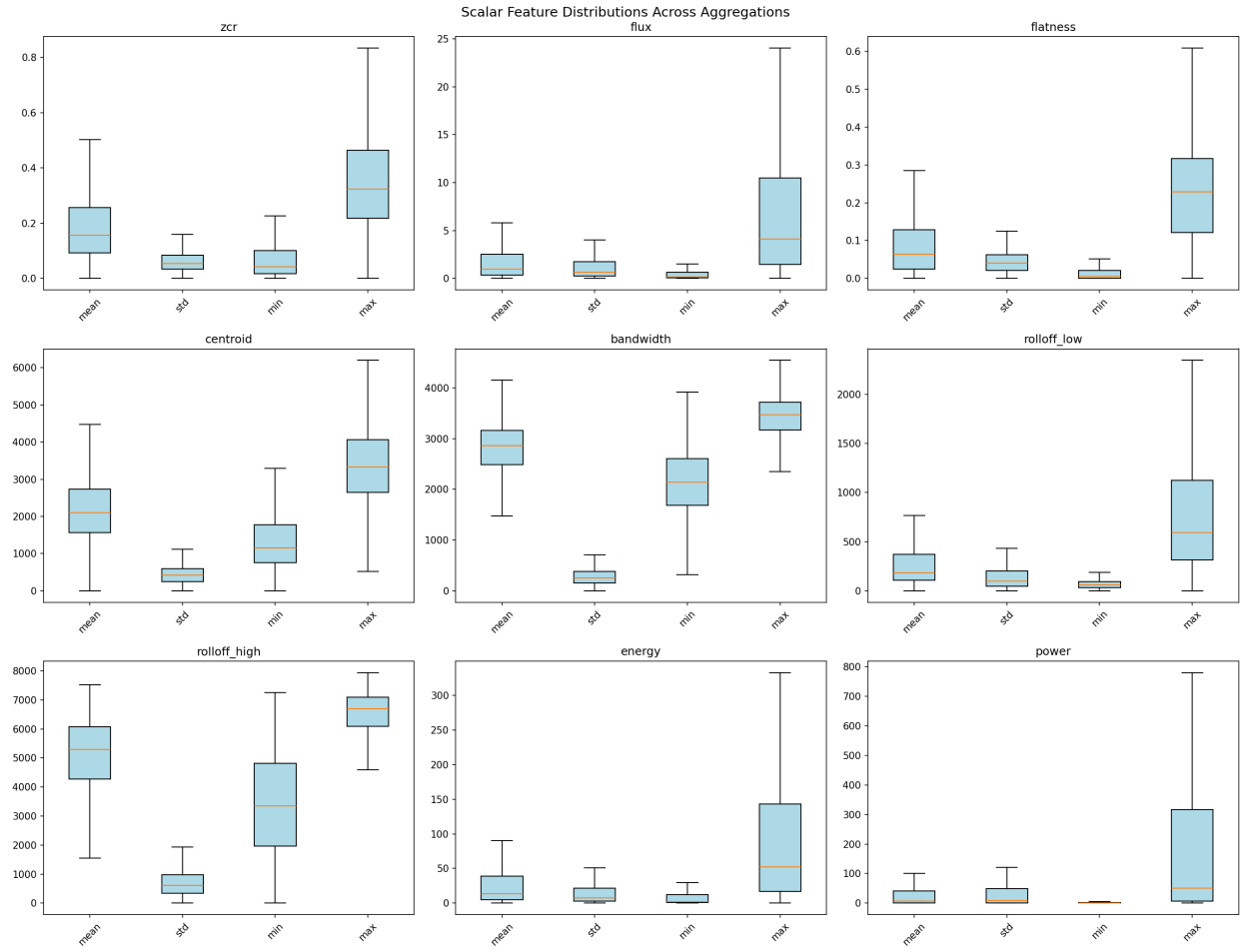


Figure 9: Value distributions for unidimensional features) across the four aggregations. Outliers are hidden for readability.

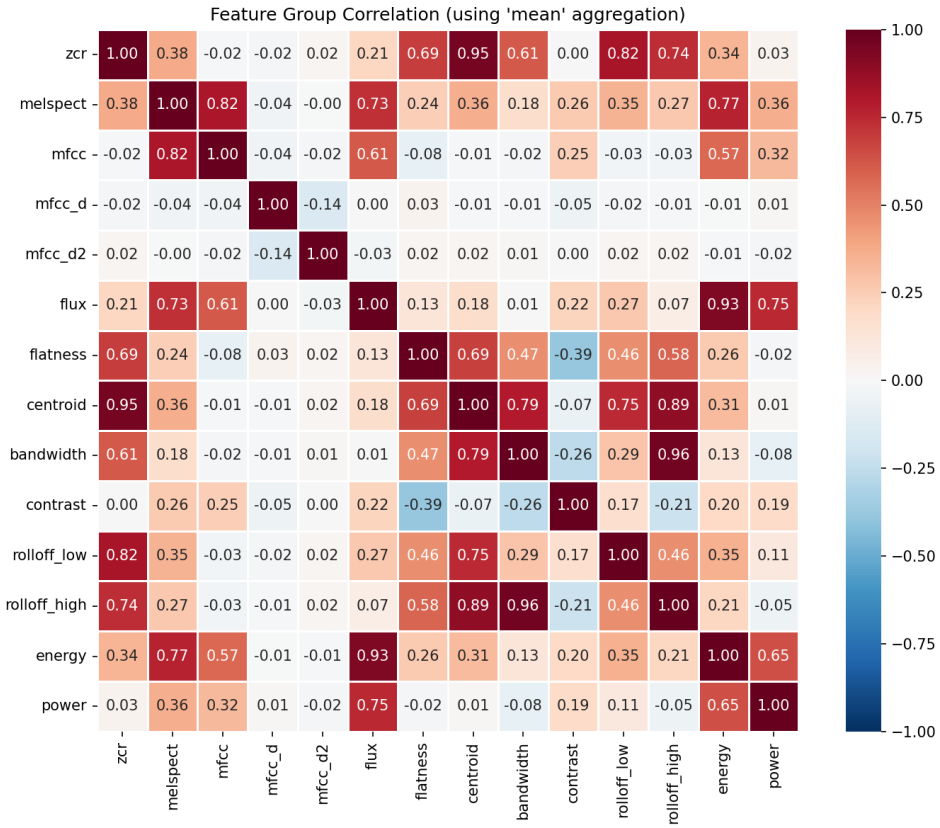


Figure 10: Feature group correlation matrix (mean aggregation).

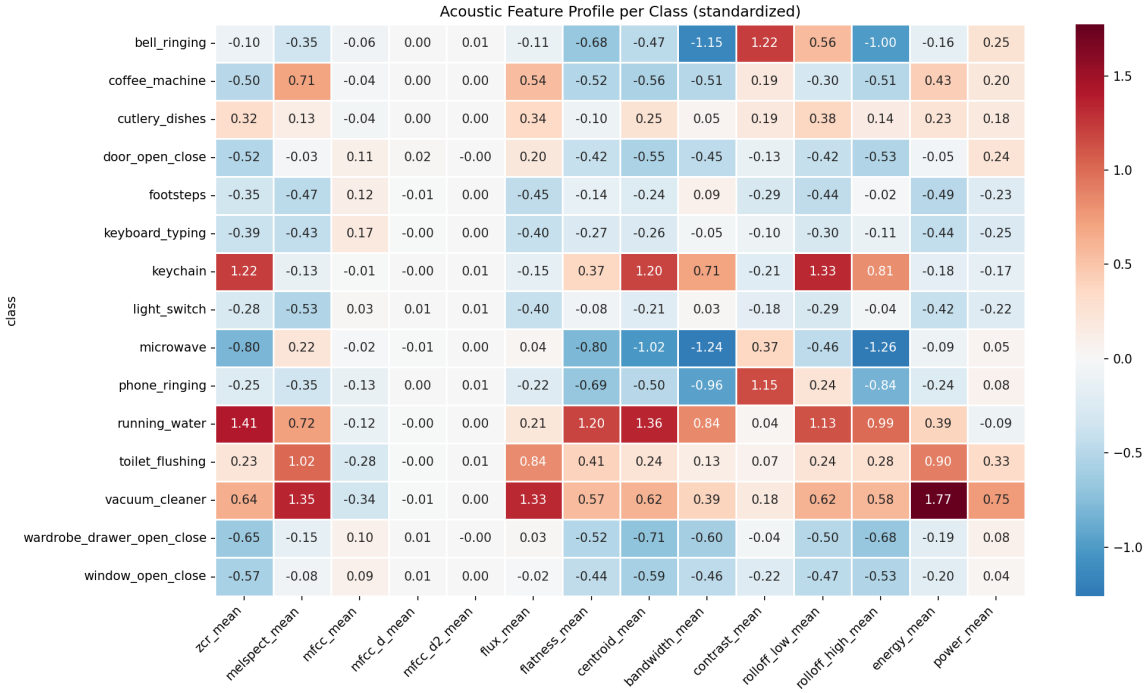


Figure 11: Acoustic feature profile per class (standardized). Each cell shows the mean deviation from the dataset average in standard deviations. Distinctive fingerprints are visible: vacuum cleaner (high energy, melspect), running water (high zcr, flatness), microwave (low centroid, bandwidth).

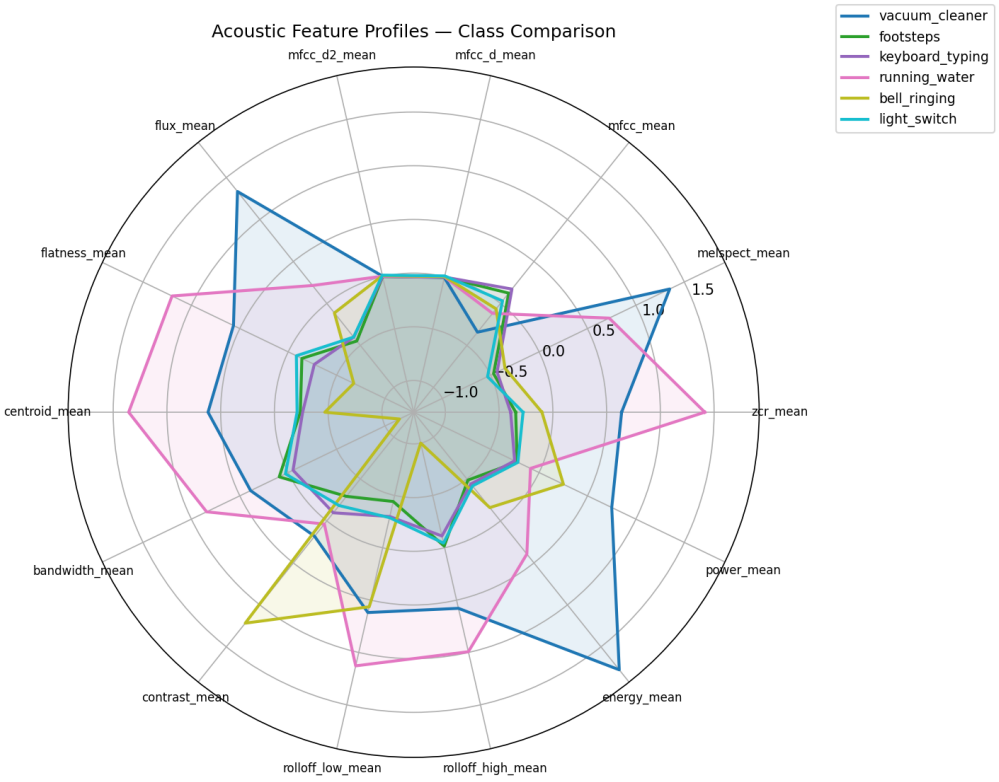


Figure 12: Radar chart comparing acoustic profiles of six selected classes. Each axis represents a mean-aggregated feature.

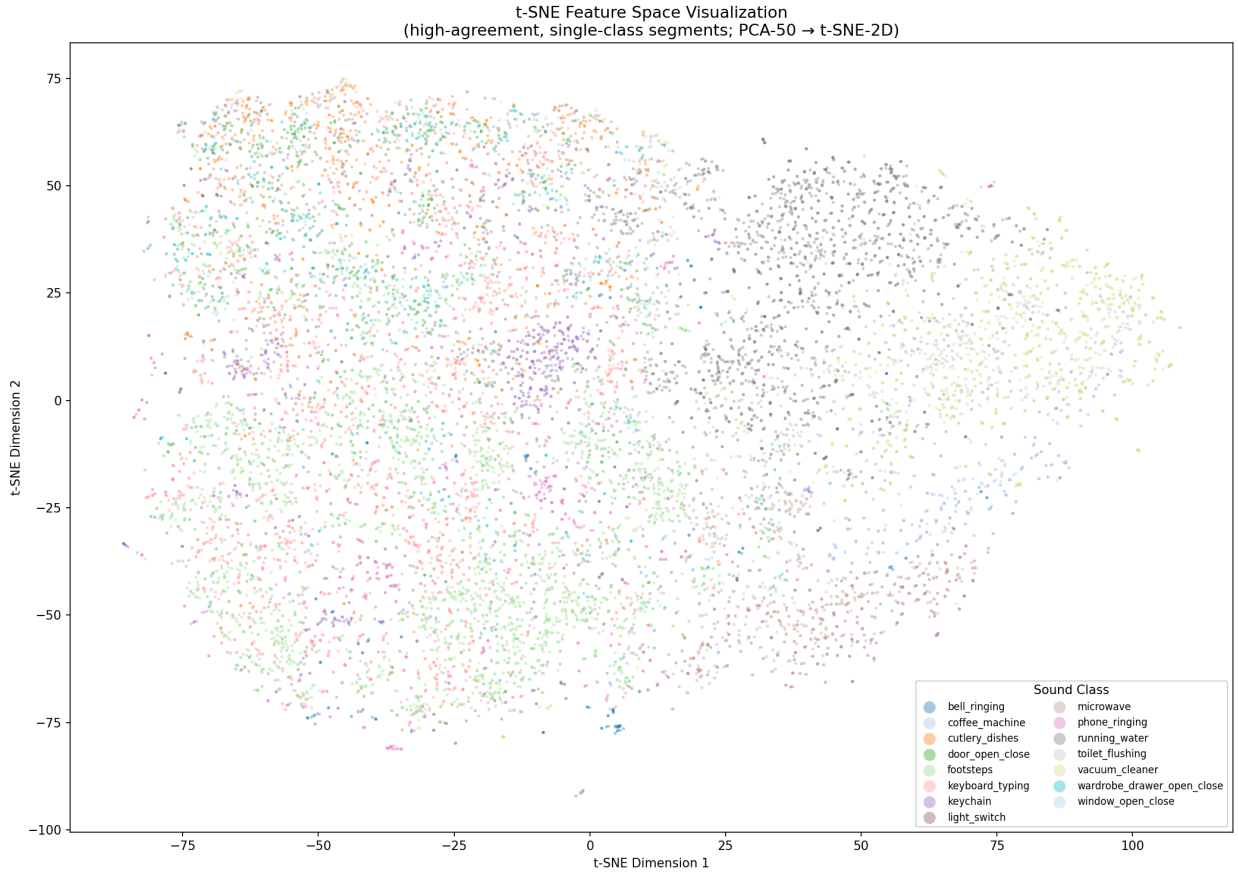


Figure 13: t-SNE visualization of spectral features (PCA-50, perplexity=30). Sustained sounds (vacuum cleaner, running water, microwave) form tight, well separated clusters. Transient events (footsteps, light switch, door) are more diffuse and overlapping.

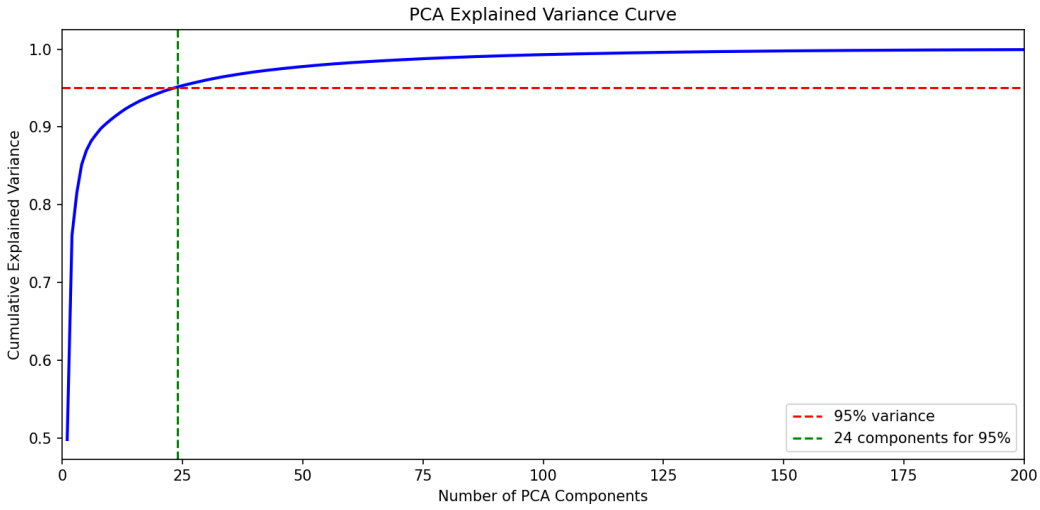


Figure 14: Cumulative PCA explained variance for spectral features (280 dims). 9 components capture 90%; 24 capture 95%; 50 capture 97.8%.

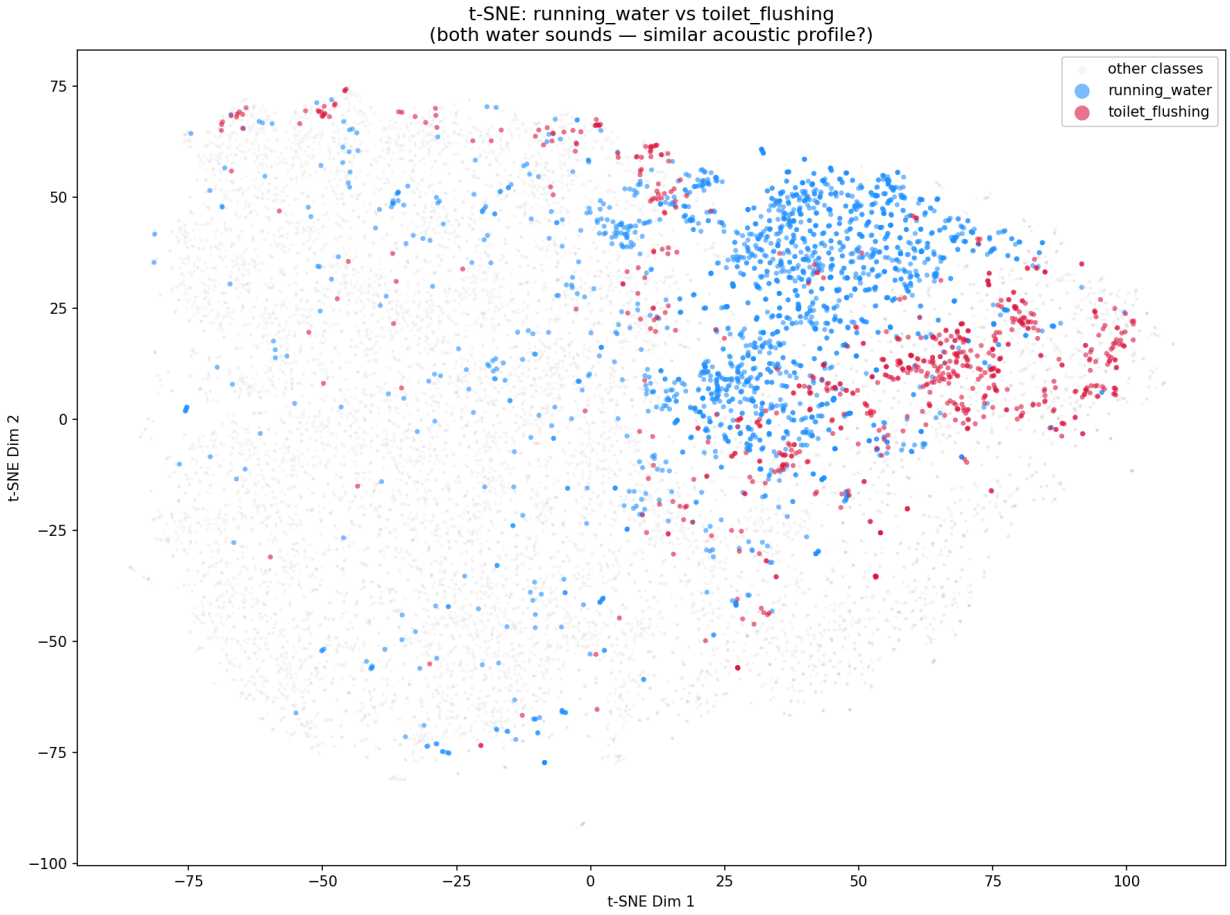


Figure 15: t-SNE pair highlight: running_water vs. toilet_flushing.

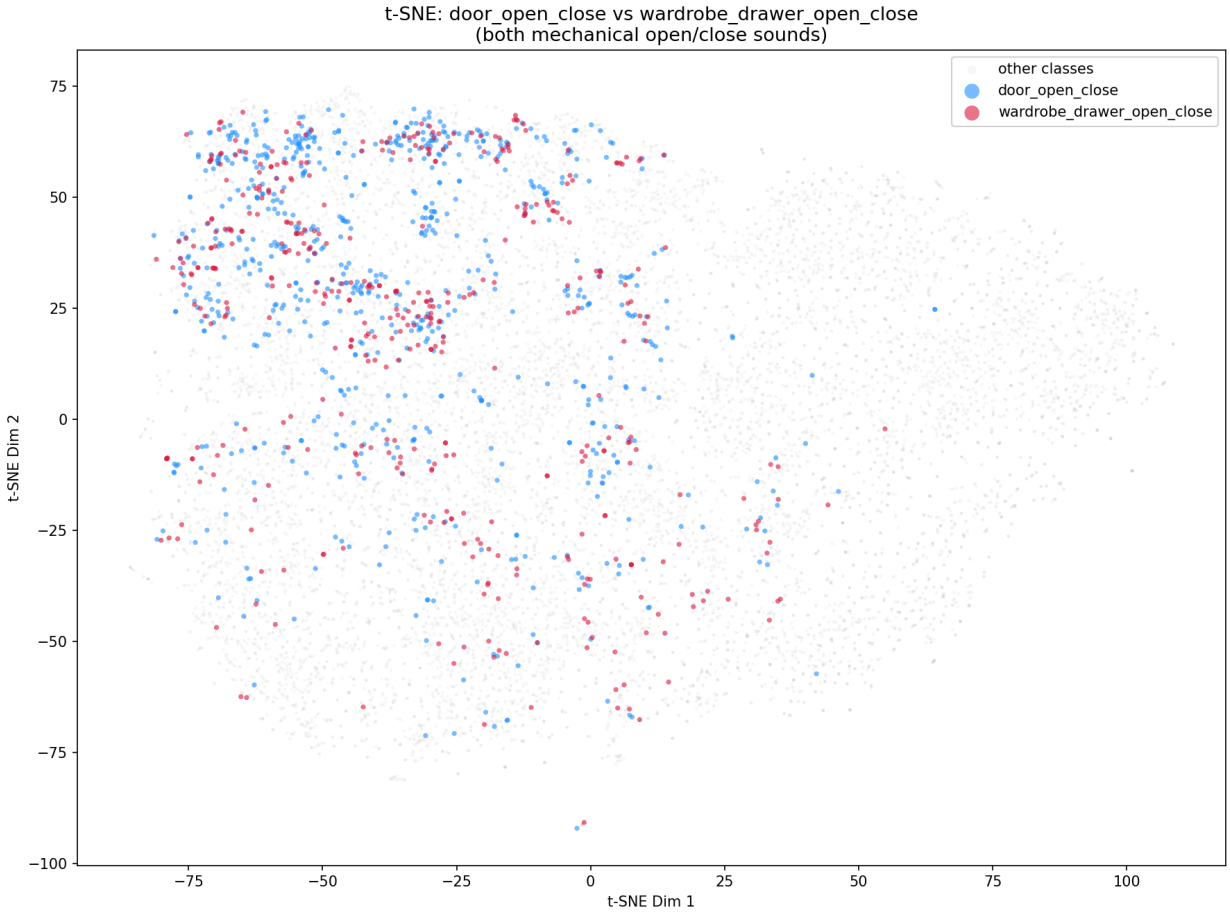


Figure 16: t-SNE pair highlight: door_open_close vs. wardrobe_drawer_open_close.

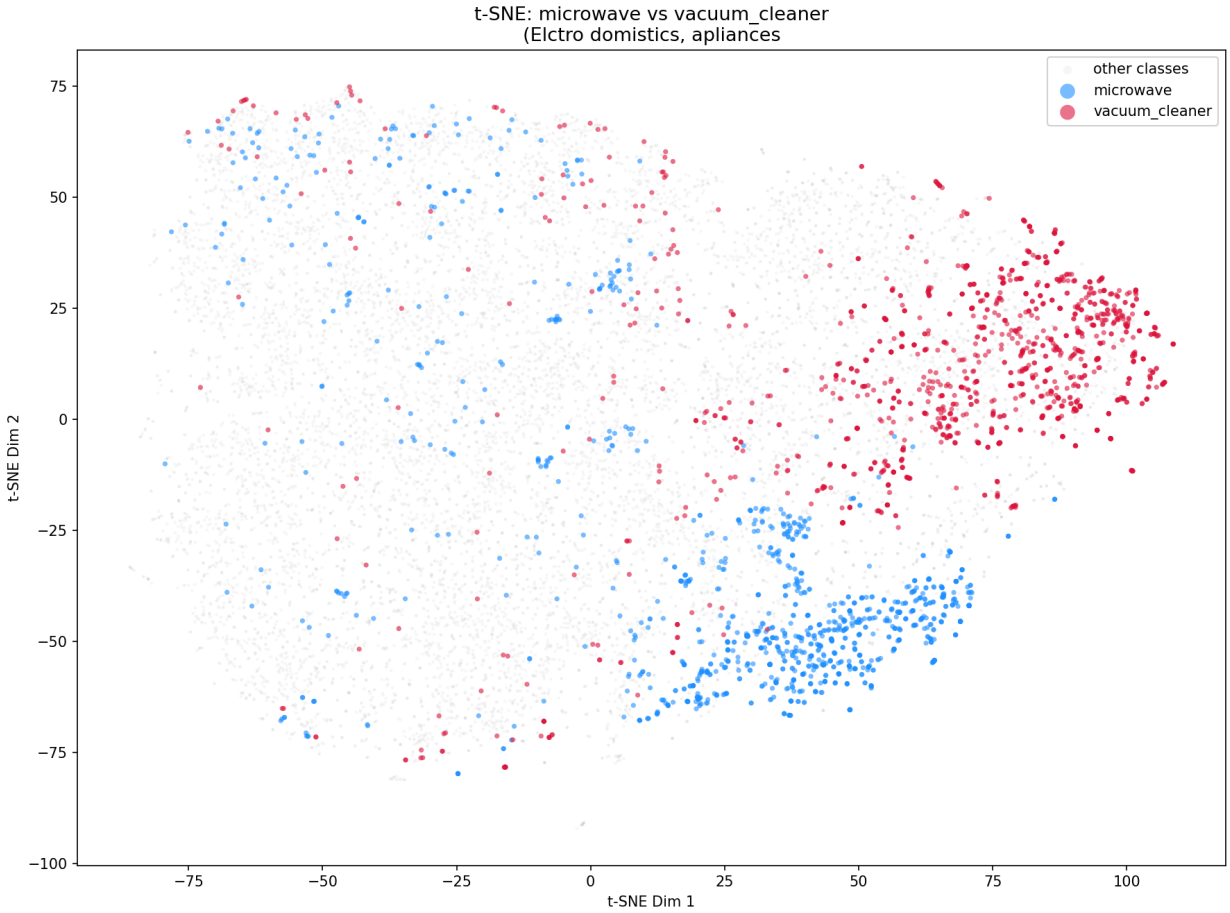


Figure 17: t-SNE pair highlight: vacuum_cleaner vs. microwave.

6.2 Tables

Table 1: Label characteristics for all 15 classes, sorted by file count.

Class	Files	Events	Med. dur (s)	Total (s)	% time
footsteps	1815	3063	7	25683	15.3
door_open_close	1155	1988	3	8347	5.0
running_water	995	1083	17	22834	13.6
cutlery_dishes	791	1681	4	11976	7.1
phone_ringing	771	1109	7	11926	7.1
keyboard_typing	725	1051	13	16867	10.0
keychain	675	1227	5	9436	5.6
wardrobe_drawer_open_close	592	1184	3	4568	2.7
window_open_close	472	698	3	2911	1.7
toilet_flushing	405	421	12	6368	3.8
microwave	390	568	24	13166	7.8
light_switch	354	430	2	1046	0.6
vacuum_cleaner	325	340	32	11452	6.8
bell_ringing	276	349	5	2541	1.5
coffee_machine	207	304	19	6898	4.1

Table 2: Most frequent sound events per recording environment (top environments)

Environment	Top 1 class	Top 2 class	Top 3 class
kitchen	running_water	cutlery_dishes	footsteps
living room	footsteps	door_open_close	keyboard_typing
office	keyboard_typing	phone_ringing	footsteps
bathroom	toilet_flushing	running_water	footsteps
bedroom	footsteps	door_open_close	wardrobe_d._o._c.
hallway	footsteps	door_open_close	keychain

Table 3: Feature statistics per feature group (raw, before normalization)

Feature group	Dims	Mean	Std	Min	Max
zcr	4	0.17	0.17	0.0	0.94
melspect	512	3.67	2.67	0.0	13.18
mfcc	128	−1.23	12.38	−130.3	44.91
mfcc_d	128	0.05	0.47	−15.4	17.21
mfcc_d2	128	0.03	0.30	−9.8	8.82
flux	4	3.19	6.87	0.0	166.60
flatness	4	0.10	0.12	0.0	1.00
centroid	4	1841.80	1349.48	0.0	7615.98
bandwidth	4	2149.69	1265.24	0.0	4951.66
contrast	28	13.83	12.99	0.0	80.11
rolloff_low	4	353.11	540.03	0.0	7437.50
rolloff_high	4	3900.31	2470.89	0.0	7937.50
energy	4	44.90	91.70	0.0	1978.07
power	4	179.04	828.71	0.0	67940.88

7 Disclosure of LLM and AI Tool Use

- **Tools Used:** Yes, we used different LLM such as Claude and DeepSeek.
- **Tasks:** The tools were used for code improvement, debugging, and generating fancy plots. Also used for brainstorming, concepts simplification, rephrasing, organizing ideas, improving clarity of technical descriptions and grammatical and orthographical checks. All generated code was executed on our dataset, outputs were manually verified against expected behavior, and all results reported in this document were produced by running the verified code on the actual dataset. Interpretations and conclusions were written by the authors based on the observed results and outputs of the implemented codes.